

English Version

Problem 2.1 The Poisson distribution and flying bombs

(a) Derive the maximum-likelihood estimate of λ

The probability mass function of the Poisson distribution is:

$$p(x = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Given i.i.d. samples $\{k_1, k_2, \dots, k_N\}$, the likelihood function is:

$$L(\lambda) = \prod_{i=1}^N \frac{\lambda^{k_i} e^{-\lambda}}{k_i!}$$

The log-likelihood function is:

$$\ell(\lambda) = \sum_{i=1}^N [k_i \log \lambda - \lambda - \log(k_i)] = \left(\sum_{i=1}^N k_i \right) \log \lambda - N\lambda - \sum_{i=1}^N \log(k_i)$$

To find the maximum likelihood estimate, differentiate with respect to λ and set to zero:

$$\frac{d\ell}{d\lambda} = \frac{\sum_{i=1}^N k_i}{\lambda} - N = 0$$

Solving for λ :

$$\hat{\lambda} = \frac{\sum_{i=1}^N k_i}{N}$$

Thus, the MLE is the sample mean.

(b) Show that the ML estimator is unbiased, and the estimator variance is $\frac{\lambda}{N}$

The estimator is $\hat{\lambda} = \frac{1}{N} \sum_{i=1}^N k_i$. The expected value is:

$$E[\hat{\lambda}] = E\left[\frac{1}{N} \sum_{i=1}^N k_i\right] = \frac{1}{N} \sum_{i=1}^N E[k_i] = \frac{1}{N} \sum_{i=1}^N \lambda = \lambda$$

So, $\hat{\lambda}$ is unbiased.

The variance is:

$$\text{Var}(\hat{\lambda}) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N k_i\right) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(k_i) = \frac{1}{N^2} \sum_{i=1}^N \lambda = \frac{\lambda}{N}$$

Thus, the variance is $\frac{\lambda}{N}$.

(c) Using the above data, calculate the ML estimate $\hat{\lambda}$ for the Poisson distribution

The data is:

- 0 hits: 229 cells
- 1 hit: 211 cells
- 2 hits: 93 cells
- 3 hits: 35 cells
- 4 hits: 7 cells
- 5+ hits: 1 cell

Total number of cells, $N = 229 + 211 + 93 + 35 + 7 + 1 = 576$.

Total number of hits, $\sum k_i = (0 \times 229) + (1 \times 211) + (2 \times 93) + (3 \times 35) + (4 \times 7) + (5 \times 1) = 0 + 211 + 186 + 105 + 28 + 5 = 535$.

Thus,

$$\hat{\lambda} = \frac{535}{576} \approx 0.9288$$

(d) Use the estimate $\hat{\lambda}$ to predict the expected number of cells with k hits, and compare with observed data

Using $\hat{\lambda} = 0.9288$, compute the Poisson probabilities:

$$p(k) = \frac{\hat{\lambda}^k e^{-\hat{\lambda}}}{k!}$$

First, $e^{-\hat{\lambda}} = e^{-0.9288} \approx 0.3950$.

Then:

- $p(0) = e^{-0.9288} \approx 0.3950$
- $p(1) = 0.9288 \times 0.3950 \approx 0.3669$
- $p(2) = \frac{0.9288^2}{2} \times 0.3950 \approx \frac{0.8627}{2} \times 0.3950 \approx 0.43135 \times 0.3950 \approx 0.1704$
- $p(3) = \frac{0.9288^3}{6} \times 0.3950 \approx \frac{0.8010}{6} \times 0.3950 \approx 0.1335 \times 0.3950 \approx 0.0527$
- $p(4) = \frac{0.9288^4}{24} \times 0.3950 \approx \frac{0.7437}{24} \times 0.3950 \approx 0.03099 \times 0.3950 \approx 0.01224$
- $p(5+) = 1 - [p(0) + p(1) + p(2) + p(3) + p(4)] \approx 1 - (0.3950 + 0.3669 + 0.1704 + 0.0527 + 0.01224) = 1 - 0.99724 = 0.00276$

Expected number of cells = $576 \times p(k)$:

- k=0: $576 \times 0.3950 \approx 227.52$
- k=1: $576 \times 0.3669 \approx 211.33$
- k=2: $576 \times 0.1704 \approx 98.15$
- k=3: $576 \times 0.0527 \approx 30.35$
- k=4: $576 \times 0.01224 \approx 7.05$
- k=5+: $576 \times 0.00276 \approx 1.59$

Observed number of cells:

- k=0: 229
- k=1: 211
- k=2: 93
- k=3: 35
- k=4: 7
- k=5+: 1

Comparison: The expected values are very close to the observed values, indicating that the Poisson distribution fits the data well. This suggests that the bomb hits were random, with no evidence of targeting specific areas.